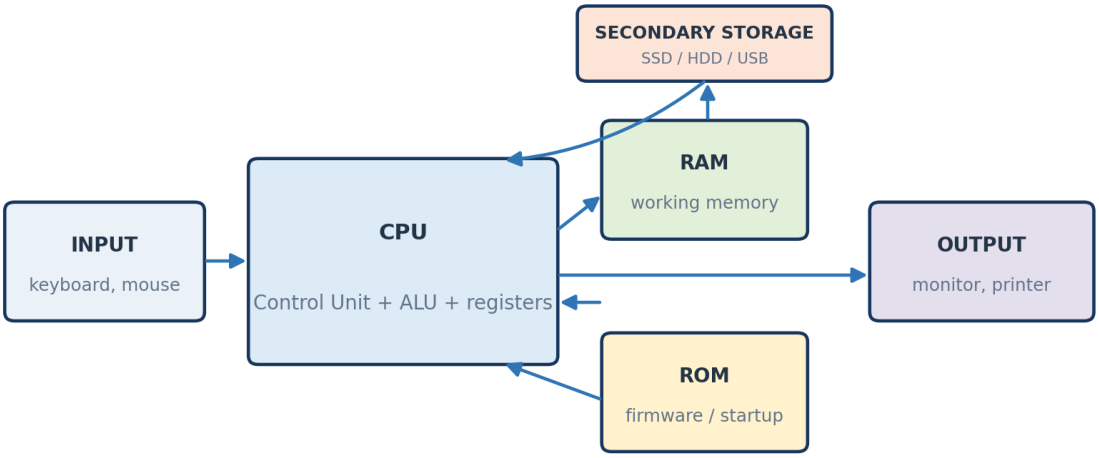
ZIMSEC

O LEVEL COMPUTER SCIENCE

CONCISE REVISION NOTES & PRACTICE QUESTIONS

Forms 1-4 | Heritage-Based Curriculum | 2024-2030



Data enters through input, is processed by the CPU, and results are sent to output or saved.

KEY IDEAS • LABELLED DIAGRAMS • EXAM TIPS • SELF-TEST

Prepared as a quick-reference study booklet for O Level candidates

HOW TO USE THIS BOOKLET

- Read each topic summary, then cover the notes and try to explain the key terms from memory.
- Practise the worked conversions, trace tables, flowcharts, database queries and code examples on paper or a computer.
- Use the diagrams to practise drawing and labelling. Examiners award marks for correct technical terms and clear labels.
- Attempt the practice questions before checking the answer guide at the end.

Remember These notes are a concise revision aid, not a replacement for the full syllabus, classroom practical work or official examination instructions.

SYLLABUS TOPIC CHECKLIST

No.	Syllabus area	What to revise
1	Hardware and software	Devices, CPU, memory, storage, operating systems, utilities, maintenance
2	Applications of Computer Science	Computing in society, intelligent systems, robotics, control systems, GIS, modelling
3	Data representation	Binary, denary, octal, hexadecimal, ASCII, storage units, binary arithmetic, logic gates
4	Communication networks and Internet technologies	Network types/topologies, media, transmission, devices, protocols, OSI/TCP-IP, cloud/mobile
5	Security and ethics	Cybercrime, privacy, integrity, validation/verification, protection, backup, ethical use
6	Systems analysis and design	Problem investigation, feasibility, SDLC, DFDs, design, testing, documentation
7	Algorithm design and problem solving	Pseudocode, flowcharts, sequence, selection, iteration, trace tables, decomposition
8	Programming	Data types, variables, operators, functions, interfaces, errors, testing and debugging
9	Databases	Tables, fields, records, keys, relationships, queries, forms, reports, SQL, security
10	Web design	HTML, website content, CMS, navigation, graphics, web security, testing
11	Technopreneurship	Innovation, market research, business plans, funding, ethics, intellectual property

EXAM PREPARATION SNAPSHOT

Paper	Focus	How to prepare
4021/1 - Multiple Choice	Broad syllabus knowledge; short factual and applied questions.	Practise definitions, comparisons, conversions, logic gates and scenario questions.
4021/2 - Structured	Explain, describe, calculate, draw, trace and write algorithms/queries.	Show working; use precise terms; label diagrams; answer the command word.
4021/3 - Practical	Programming, database and web design tasks in the 2026 specimen format.	Practise coding, database queries and web pages.

Exam tip For a question worth 3 marks asking for three differences or advantages, give three separate, distinct points. Do not repeat the same idea in different words.

1. HARDWARE AND SOFTWARE

Know what each computer component does and select suitable devices for a given task.

1.1 Essential terms

Term	Meaning
Hardware	Physical parts of a computer that can be touched.
Software	Programs/instructions that tell hardware what to do.
Data	Raw facts entered into a computer.
Information	Processed data that is meaningful to the user.
Peripheral	A device connected to a computer, such as a printer or scanner.
Firmware	Software stored in non-volatile memory that controls a hardware device.

1.2 Main hardware groups

Group	Examples	Purpose
Input	Keyboard, mouse, scanner, microphone, barcode reader, touchscreen, sensors	Enter data or instructions.
Output	Monitor, printer, speakers, projector, plotter	Present processed information.
Processing	CPU, GPU	Carry out instructions and process data.
Primary memory	RAM, ROM, cache, registers	Hold instructions/data currently used by the CPU.
Secondary storage	HDD, SSD, USB flash drive, memory card, optical disc	Keep programs and files for long-term use.

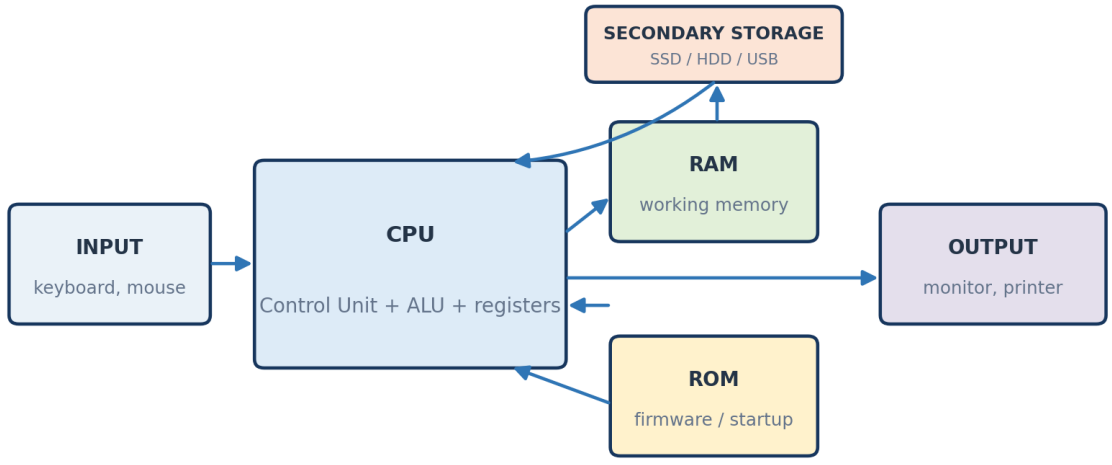


Figure 1: Simplified computer system. The CPU processes data; RAM is temporary working memory and secondary storage keeps files when power is off.

1.3 CPU and memory

- **CPU (Central Processing Unit):** executes instructions and controls computer operations.
- **ALU (Arithmetic Logic Unit):** performs arithmetic calculations and logical comparisons.
- **Control Unit (CU):** fetches and decodes instructions, then coordinates their execution.
- **Registers:** very small, very fast storage locations inside the CPU. Cache stores frequently needed data/instructions close to the CPU.
- **RAM:** volatile, read/write memory. Its contents are lost when power is switched off.
- **ROM:** non-volatile memory that stores firmware/start-up instructions.

1.4 Storage devices - quick comparison

Device	Main strengths	Limitations / suitable use
HDD	Large capacity; low cost per GB.	Moving parts; slower and more easily damaged by shock. Useful for bulk storage.
SSD	Very fast access; silent; durable because it has no moving parts.	Usually costs more per GB. Useful for operating systems and fast-loading applications.
USB flash drive / memory card	Portable, reusable and convenient.	Easy to lose; limited write life; can be infected by malware.
Optical disc	Removable; useful for distribution/archiving.	Lower capacity and slower access than many modern drives.

1.5 Software

Type	Examples	Use
System software	Operating systems, device drivers, utilities	Manages hardware and provides a platform for applications.
Application software	Word processor, spreadsheet, database, browser, graphics editor	Helps users perform specific tasks.
Utility software	Antivirus, backup, compression, disk clean-up, encryption	Maintains, protects or optimises a computer.

- Operating system functions: user interface, memory management, processor scheduling, file management, device management, security and networking.
- Off-the-shelf software is ready-made for many users; bespoke/customised software is developed or adapted for a particular customer.
- Open-source software makes its source code available under a licence that permits specified use, study or modification.

1.6 Maintenance and troubleshooting

- **Preventive maintenance:** update software, scan for malware, back up files, clean vents, check cables and use safe shutdown procedures.
- **Troubleshooting:** identify the symptom, check simple causes, test one possible cause at a time, apply a fix and confirm the system works.

Common exam distinction RAM is temporary/volatile; ROM is non-volatile. An operating system is system software; a word processor is application software.

2. APPLICATIONS OF COMPUTER SCIENCE

Be ready to link a computer system, sensor or application to a real-life problem.

Area	Examples of computer use
Agriculture	Automated irrigation, soil/moisture monitoring, weather forecasting, farm records, crop monitoring.
Banking	ATMs, online banking, point-of-sale systems, electronic funds transfer and fraud detection.
Education	E-learning, digital libraries, learner records, simulations and computer-based assessment.
Health	Patient records, diagnostic imaging, appointment systems and patient monitoring.
Transport	Traffic control, GPS navigation, ticketing, fleet tracking and automated tolling.
Mining/manufacturing	Computer-aided design (CAD), computer-aided manufacturing (CAM), machine monitoring and robotics.
Environment/wildlife	GIS mapping, satellite images, animal tracking, pollution monitoring and early warning systems.

2.1 Important concepts

- **GIS (Geographic Information System):** captures, stores, analyses and displays data linked to locations. Used for maps, land planning, wildlife tracking and resource management.
- **CAD:** uses software to create or modify a design. CAM: uses computers to control manufacturing equipment.
- Modelling: creating a simplified representation of a real system. Simulation: running the model to investigate how the system behaves under different conditions.
- **Intelligent system:** uses data, rules or learning methods to make decisions or recommendations. Examples include route-planning systems and some diagnostic tools.
- **Robotics:** designing, building and programming machines that carry out tasks. Robots may use sensors, controllers and actuators.
- A sensor detects a physical condition (e.g. temperature, light, moisture or movement) and converts it into a signal a system can process.

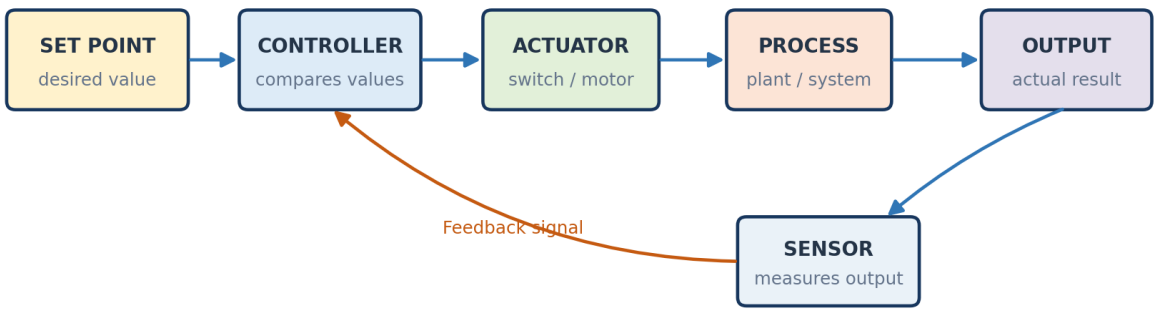


Figure 2: Closed-loop control system. Feedback allows the controller to compare the measured output with the desired set point.

Control-system example A greenhouse system reads temperature from a sensor. The controller compares it with the target and switches a fan on or off through an actuator.

2.2 Advantages and limitations

- **Advantages:** speed, accuracy, consistency, 24-hour operation, safer work and better decisions.
- **Limitations:** cost, maintenance, possible job displacement, dependence on electricity/data and failure or inaccurate-input risks.

3. DATA REPRESENTATION

Computers store and process data in binary. Practise conversions and show every step.

3.1 Units of data

Unit	Meaning
Bit	A single binary digit: 0 or 1.
Nibble	4 bits.
Byte	8 bits.
Kilobyte (KB)	Traditionally 1,024 bytes in binary-based school calculations.
Megabyte (MB)	1,024 KB.
Gigabyte (GB)	1,024 MB.
Terabyte (TB)	1,024 GB.

3.2 Number systems

Number system	Base	Digits used
Binary	2	0, 1
Octal	8	0 to 7
Denary	10	0 to 9
Hexadecimal	16	0 to 9 and A to F

Binary place values From right to left, binary column values are 1, 2, 4, 8, 16, 32, 64, 128 ... (powers of 2).

3.3 Worked conversions

- Binary to denary: $101101_2 = 32 + 0 + 8 + 4 + 0 + 1 = 45_{10}$.
- Denary to binary: divide by 2 repeatedly and read the remainders from bottom to top. Example: $45_{10} = 101101_2$.
- Binary to octal: group bits in threes from the right. $111010_2 \rightarrow 111\ 010 \rightarrow 7\ 2 \rightarrow 72_8$.
- Binary to hexadecimal: group bits in fours from the right. $10101101_2 \rightarrow 1010\ 1101 \rightarrow A\ D \rightarrow AD_{16}$.
- Hexadecimal to binary: replace each hex digit with four bits. A = 1010; D = 1101; therefore $AD_{16} = 10101101_2$.

3.4 Binary arithmetic

Addition rules	Subtraction rules
0 + 0 = 0	0 - 0 = 0
0 + 1 = 1	1 - 0 = 1
1 + 0 = 1	1 - 1 = 0
1 + 1 = 10 (write 0, carry 1)	0 - 1 requires borrowing from the next column

- Example addition: $1011_2 + 0110_2 = 10001_2$ (11 + 6 = 17).
- Example subtraction: $10110_2 - 01001_2 = 01101_2$ (22 - 9 = 13).

3.5 Character representation

- ASCII represents characters using numeric codes. Standard ASCII uses 7 bits (128 codes); extended versions use 8 bits. Unicode supports a much wider range of characters and writing systems.

3.6 Logic gates

- AND outputs 1 only when all inputs are 1. OR outputs 1 when at least one input is 1. NOT reverses its input.
- NAND is the inverse of AND; NOR is the inverse of OR. XOR outputs 1 when its two inputs are different. Practise truth tables for all six gates.

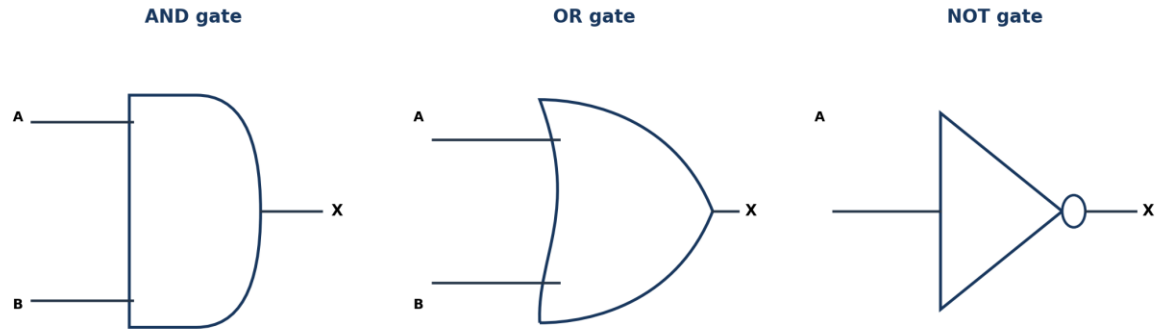


Figure 3: Simplified gate illustrations. In circuit questions, follow the exact standard symbols shown in class or the exam reference.

A	B	AND	OR	XOR
0	0	0	0	0
0	1	0	1	1
1	0	0	1	1
1	1	1	1	0

Revision drill Memorise the place values, practise each conversion direction, and always show your working. In a logic truth table, list every possible input combination.

4. COMMUNICATION NETWORKS AND INTERNET TECHNOLOGIES

Focus on why a network is used, how devices are connected and how data travels.

4.1 Network types

Type	Coverage / example
PAN	Personal Area Network: a few metres around one person, e.g. phone connected to earbuds by Bluetooth.
LAN	Local Area Network: a home, office, school or single building.
MAN	Metropolitan Area Network: connects sites across a town or city.
WAN	Wide Area Network: covers large geographical areas; the Internet is the best-known example.

4.2 Network topologies

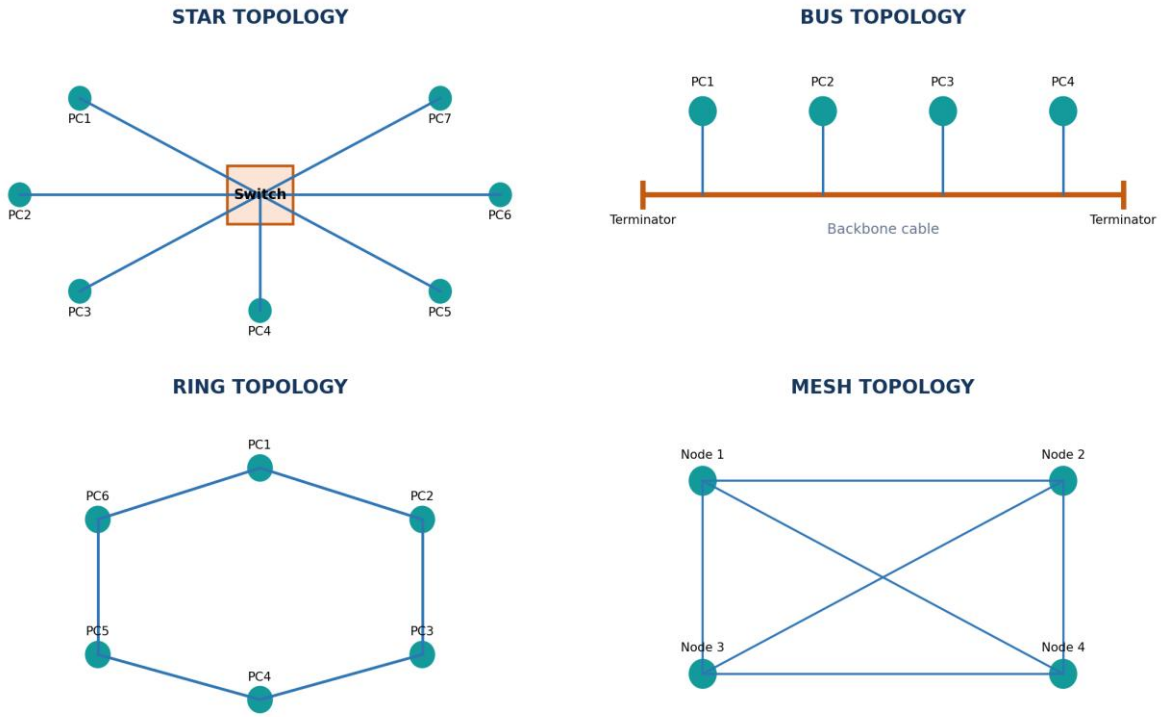


Figure 4: Common topologies. Practise drawing each with devices and connection lines clearly labelled.

Topology	Strength	Weakness
Star	Easy to add/manage devices; one cable failure usually affects one device only.	Central switch/hub failure can stop the whole network; more cabling.
Bus	Simple and inexpensive for a small network.	Backbone failure affects all devices; performance falls with traffic.
Ring	Each device has a predictable path for data.	A break or failed device can disrupt the ring unless a bypass exists.
Mesh	Multiple paths; high reliability and fault tolerance.	Expensive and complex because many links are required.

4.3 Transmission modes



Figure 5: Simplex is one-way; half-duplex is two-way but alternating; full-duplex is two-way simultaneously.

4.4 Transmission media

Medium	Characteristics
Twisted-pair copper	Low cost and easy to install; common in Ethernet; can be affected by electromagnetic interference.
Coaxial cable	Shielded copper cable; better protection from interference than basic twisted pair; used in some broadband/TV systems.
Optical fibre	Uses light; very high speed and long distance; immune to electromagnetic interference; installation/repair may cost more.
Wireless	Uses radio/microwave/infrared signals; mobile and convenient; affected by distance, obstacles, interference and security risks.

4.5 Network devices

Device	Function
NIC	Connects a computer/device to a network; provides a network interface and hardware address.
Hub	Repeats incoming data to all ports; does not intelligently select the destination.
Switch	Forwards frames to the appropriate LAN port using MAC addresses.
Router	Connects different networks and forwards packets using IP addresses.
Bridge	Connects or segments LAN sections and filters traffic by MAC address.
Modem	Modulates/demodulates signals to connect to an ISP over a communication service.
Wireless access point	Allows wireless devices to connect to a wired network.

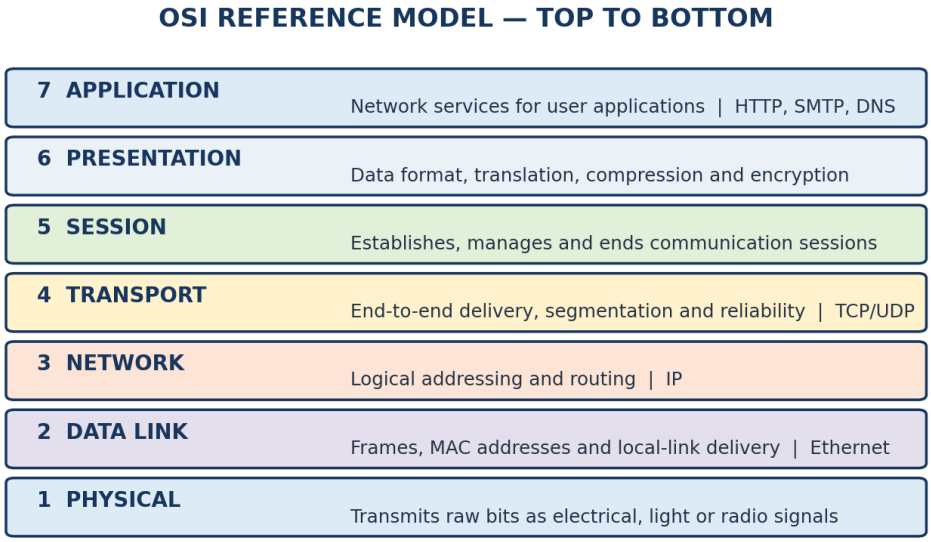
4.6 Protocols and Internet services

Protocol / service	Main purpose
TCP/IP	Core protocol suite used to address and transport data across networks.
HTTP / HTTPS	Transfers web pages; HTTPS adds encryption and authentication using TLS.
FTP	Transfers files between computers.
SMTP	Sends email between mail servers / from clients to servers.
POP3 / IMAP	Retrieves email; IMAP keeps mail synchronised on the server.
DNS	Translates a domain name into an IP address.
DHCP	Automatically leases IP configuration to devices.
ISP	Company that provides Internet access and related services.

4.7 Models, mobile technology and cloud

- OSI model has seven layers: Physical, Data Link, Network, Transport, Session, Presentation and Application. Remember from bottom to top: Please Do Not Throw Sausage Pizza Away.

- TCP/IP model is commonly shown as Network Access, Internet, Transport and Application. It is the practical protocol suite used on the Internet.
- Wireless/mobile technologies include Wi-Fi, Bluetooth, WiMAX, radio, microwave, infrared, satellite, mobile phones and tablets.
- Cloud computing provides services over a network. Examples: online storage, web applications and hosted computing resources. Benefits include access from different locations and scalable capacity; risks include Internet dependence, privacy and provider outages.



Mnemonic, from layer 1 up to layer 7: Please Do Not Throw Sausage Pizza Away.

Figure 4: The OSI model. Layer 1 is nearest the physical medium; layer 7 is nearest the user application.

Exam comparison A hub broadcasts to every port; a switch forwards within a LAN to the intended port; a router connects different networks.

5. SECURITY AND ETHICS

Protect information, systems and people. Apply the ideas to realistic scenarios.

5.1 Security objectives

Objective	Meaning	Example protection
Confidentiality	Only authorised people can see data.	Passwords, access rights, encryption.
Integrity	Data remains accurate and is not altered improperly.	Validation, verification, hashes, permissions.
Availability	Systems/data are accessible when required.	Backups, UPS, redundancy, disaster recovery.

5.2 Threats and crimes

Threat / crime	Meaning
Malware	Malicious software such as viruses, worms, Trojans, spyware or ransomware.
Phishing	Fake messages or websites trick users into revealing confidential information.
Hacking / unauthorised access	Accessing a computer or account without permission.
Spoofing	Pretending to be a trusted person, device, email address or website.
Sniffing	Capturing network traffic to inspect data; can be misused to steal information.
Identity theft / fraud	Using another person's details dishonestly for gain.
Piracy	Unauthorised copying or distribution of copyrighted software or content.

5.3 Protection measures

- Use long, unique passwords or passphrases; enable multi-factor authentication where available; never share one-time codes.
- Use firewalls to control network traffic and anti-malware software to detect/remove malicious programs.
- Encrypt data so intercepted or stolen files are difficult to read without the key.
- Apply user accounts, file permissions and least-privilege access. Lock devices and keep software patched.
- Back up important data regularly and test that backups can be restored.

5.4 Verification and validation

Term	Meaning	Example
Validation	Checks whether input is sensible/within rules.	A mark field accepts 0 to 100; a date must be in a valid range.
Verification	Checks that data was copied or entered exactly as intended.	Double-entry checking; visual comparison with the source.
Authentication	Checks who a user is.	Password, PIN, fingerprint or security token.
Authorisation	Determines what an authenticated user is allowed to do.	A learner can view marks but only a teacher can edit them.

5.5 Backup and disaster recovery

- Full backup: copies all selected data. Incremental backup: copies changes since the last backup of any type. Differential backup: copies changes since the last full backup.
- A disaster recovery plan states how an organisation will restore data, systems and essential services after a major incident.
- Keep backups separate, use multiple locations and test recovery. Respect privacy and copyright, avoid plagiarism/piracy and practise safe, balanced online behaviour.

6. SYSTEMS ANALYSIS AND DESIGN

The systems development life cycle (SDLC) provides a structured way to build or improve an information system.

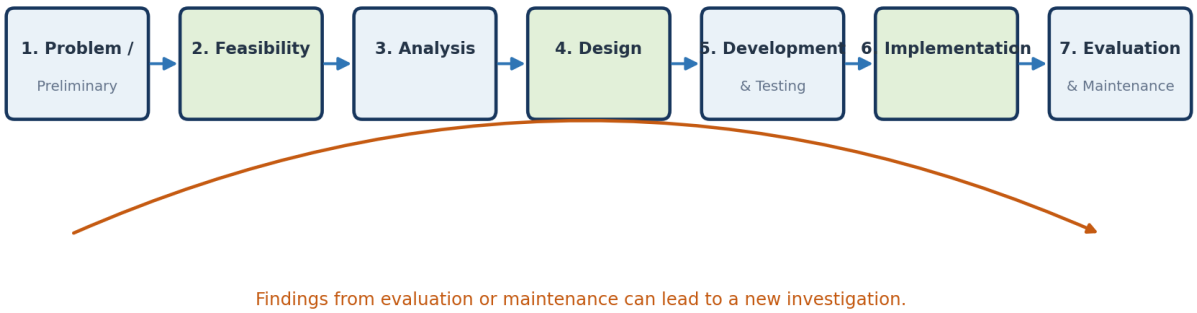


Figure 6: A simplified SDLC. Iteration may occur when testing or evaluation identifies a problem.

Stage	Main activities
1. Preliminary investigation / problem identification	Understand the problem, current system, objectives, scope and constraints.
2. Feasibility study	Check whether the proposed solution is practical: technical, economic, legal, operational and social feasibility.
3. Systems analysis	Gather requirements; study inputs, processes, outputs, data and user needs; consider alternative solutions.
4. Systems design	Plan inputs, outputs, user interface, files/database, algorithms, security and network requirements.
5. Development and testing	Write code/build components; test modules and the complete system; correct errors.
6. Implementation	Install the system, convert data, migrate users and put the system into operation.
7. Documentation and user training	Prepare technical/user guides and train staff to operate the new system.
8. Evaluation and maintenance	Review against objectives; correct faults, adapt to changes and improve the system.

6.1 Data collection techniques

Technique	Use / limitation
Questionnaire	Collects standard responses from many users; questions must be clear.
Interview	Allows detailed answers and follow-up questions; takes time.
Observation	Analyst watches the current work being done; users may change behaviour when observed.
Record / document inspection	Examines existing forms, reports and files; may reveal missing or duplicated data.

6.2 Feasibility - TELOS + social

- Technical: do we have the hardware, software, skills and infrastructure?
- Economic: are the benefits worth the development and running costs?
- Legal: does it comply with relevant laws, licences and data-protection requirements?
- Operational: will it work in the organisation and be accepted by users?
- Social: what effect will it have on people, communities and working practices?

6.3 Design tools

- Data Flow Diagram (DFD): shows how data moves between external entities, processes and data stores.
- System flowchart: shows the sequence of operations, inputs, outputs, storage and processing in a system.
- Activity diagram: models activities and the order/flow of work in a process.

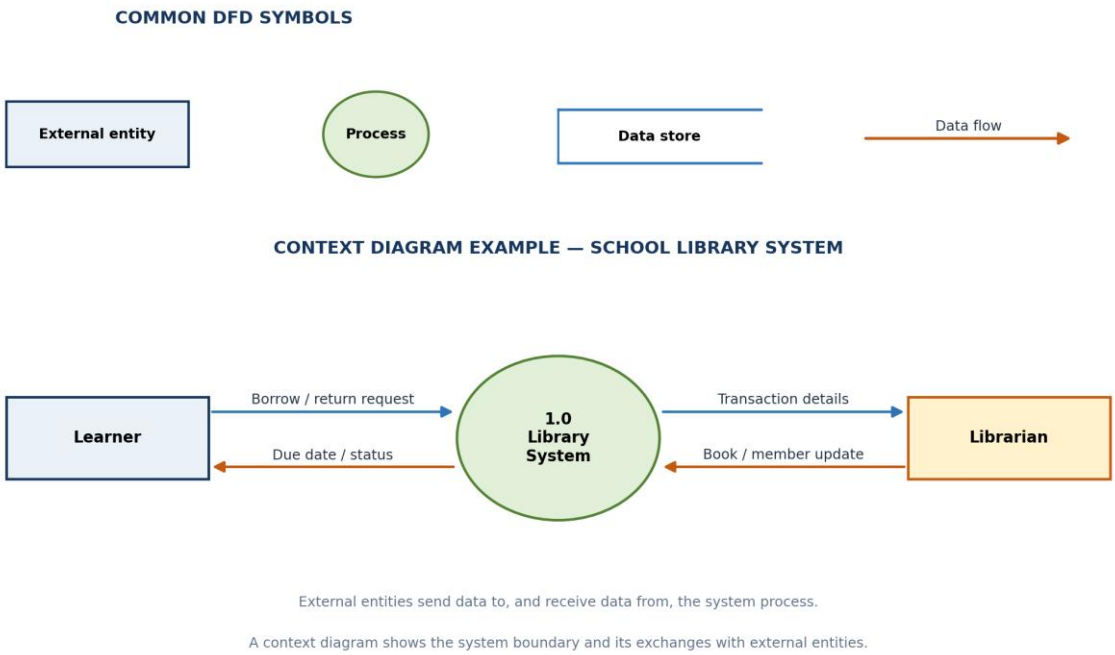


Figure 7: Common DFD symbols and a simple context diagram for a library system.

Exam tip Do not confuse analysis with design. Analysis asks what the system must do; design specifies how the proposed system will do it.

7. ALGORITHM DESIGN AND PROBLEM SOLVING

An algorithm is a finite, ordered set of unambiguous steps for solving a problem.

7.1 Good algorithm characteristics

- Has clearly defined inputs and outputs, unambiguous instructions, a finite number of steps and produces the intended result.
- Decomposition breaks a large problem into smaller manageable tasks. Top-down design starts with the whole problem and refines it into modules.
- Pseudocode describes program logic in structured English. A flowchart represents the logic using standard symbols.

7.2 Three basic control structures

Structure	Purpose	Typical pseudocode
Sequence	Steps run one after another.	READ mark percentage = mark * 4 DISPLAY percentage
Selection	Chooses a path based on a condition.	IF mark >= 50 THEN DISPLAY "Pass" ELSE DISPLAY "Fail" ENDIF
Iteration / repetition	Repeats steps while/until a condition is met or for a set number of times.	FOR count = 1 TO 5 DISPLAY count NEXT count

7.3 Common flowchart symbols

Symbol shape	Meaning
Oval / rounded terminator	Start or end.
Rectangle	Process / calculation / assignment.
Parallelogram	Input or output.
Diamond	Decision; branches such as Yes/No.
Arrow	Direction of control flow.
Small circle	Connector between flowchart parts.

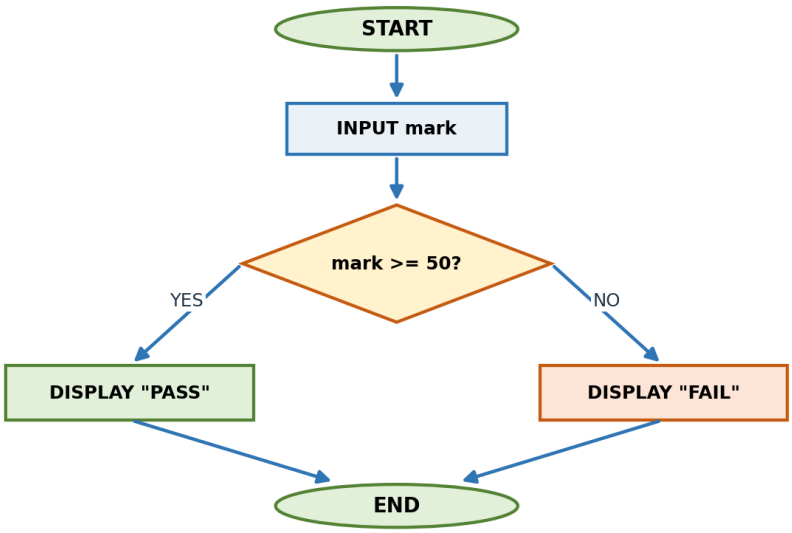


Figure 7: Example decision flowchart. Follow the Yes and No branches carefully.

7.4 Trace tables and dry runs

- A trace table records how variable values and outputs change as each algorithm statement is executed.
- Trace the algorithm in order. Update a variable only when an assignment occurs. Record each loop iteration and condition result.

Step	count	total	Output
Initial	0	0	-
1	1	1	-
2	2	3	-
3	3	6	-
End	3	6	Total = 6

Common mistakes Check comparison operators (=, <, >, <=, >=), loop start/end values, variable initialisation and whether the loop should run while or until a condition is true.

8. PROGRAMMING

Programming turns an algorithm into instructions a computer can execute.

8.1 Core terms

Term	Meaning / example
Program	A set of coded instructions that performs a task.
Variable	A named memory location whose value can change, e.g. total = 0.
Constant	A named value that should not change while the program runs, e.g. TAX_RATE = 0.15.
Data type	Defines the kind of value held: integer, real/float, character, string or Boolean.
Operator	A symbol/keyword used for arithmetic, comparison or logic.
Function	A named block of code that can return a value.
Procedure	A named block of code that performs a task; it may not return a value.

8.2 Operators

Category	Examples	Purpose
Arithmetic	+ - * / MOD	Calculations; MOD returns a remainder.
Relational	= <> < > <= >=	Compare values; result is True or False.
Logical	AND OR NOT	Combine or reverse Boolean conditions.

8.3 Control structures

- Sequence: statements execute in order.
- Selection: IF...ELSE or CASE/SELECT chooses between alternatives.
- Iteration: FOR repeats a known number of times; WHILE tests before each pass; REPEAT...UNTIL tests after a pass.

8.4 Example pseudocode

```
START
  READ learnerName
  READ mark
  percentage = (mark / 25) * 100
  IF mark >= 13 THEN
      result = "Pass"
  ELSE
      result = "Refer"
  ENDIF
  DISPLAY learnerName, percentage, result
STOP
```

8.5 Errors, testing and debugging

Error type	Meaning	Example
Syntax	Breaks the language grammar.	Missing bracket, misspelled keyword or missing quotation mark.
Runtime	Occurs while the program is running.	Dividing by zero, invalid file access or type conversion failure.
Logic	Program runs but produces a wrong result.	Using + instead of * in a total calculation.

- Testing checks whether the program behaves as required. Use normal, boundary and invalid/abnormal test data.
- Unit testing checks an individual module; integration testing checks modules working together; system testing checks the complete system; user acceptance testing checks whether it meets user needs.
- Debugging is locating and correcting errors. Retest after making a change.

8.6 User interface basics

- Common interface objects: labels, text boxes, buttons, list boxes, combo boxes, check boxes, radio buttons and menus.

- Visual programming uses forms, controls and drag-and-drop components; set each component's properties and write event code (for example, a button-click action).
- A good interface is consistent, clear, accessible, easy to navigate and gives useful feedback. Validate input before processing.

Practical reminder Practise writing code that accepts input, calculates a result, uses a decision, repeats a task and displays formatted output. Test the program with at least three different sets of data.

8.7 Example test data

- For a mark program that accepts values from 0 to 25 and awards a Pass at 13 or above, use normal, boundary and invalid test data.

Test type	Input mark	Expected result
Normal	9	Refer
Normal	18	Pass
Boundary - lower	0	Refer
Boundary - upper	25	Pass
Invalid	-1	Reject as outside range
Invalid	26	Reject as outside range

9. DATABASES

A database organises related data so it can be stored, searched, updated and reported efficiently.

9.1 Database vocabulary

Term	Meaning
Database	An organised collection of related data.
DBMS	Database Management System: software used to create, store, maintain and query databases.
Table	Stores data about one entity/type of thing.
Field	One attribute/column, e.g. StudentName.
Record	One complete row about one entity, e.g. one student.
Primary key (PK)	Field or combination of fields that uniquely identifies each record.
Foreign key (FK)	Field that refers to a key in another table to create a relationship.
Query	A request to search, filter, calculate or change data.

9.2 Database objects

Object	Purpose
Table / datasheet	Stores and displays records in rows and fields in columns.
Form	Provides a user-friendly way to enter, view or edit records.
Query	Selects or changes records that meet specified conditions.
Report	Formats data for presentation or printing.

9.3 Relationships and keys

- One-to-one: one record in Table A is linked to one record in Table B.
- One-to-many: one record in Table A can link to many records in Table B. This is common in school databases.
- Many-to-many: many records on each side can relate; use a junction/link table to resolve it into two one-to-many relationships.

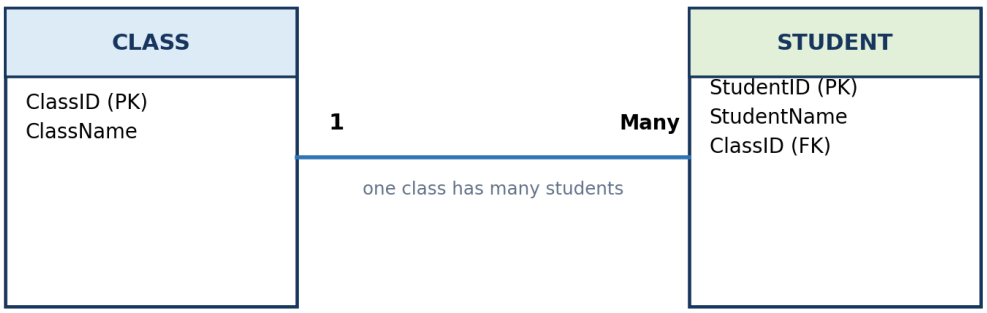


Figure 8: One class can contain many students. ClassID is the primary key in CLASS and a foreign key in STUDENT.

9.4 Data manipulation and SQL

- CRUD means Create, Read, Update and Delete. These are the four basic data operations.

```
SELECT StudentName, Mark
FROM Students
WHERE Mark >= 50
ORDER BY StudentName ASC;
```

- SELECT chooses fields; FROM identifies the table; WHERE filters records; ORDER BY sorts the result.

- Advanced queries can combine criteria with AND/OR, sort data, calculate fields, summarise with SUM/AVG/COUNT and join related tables.

```
UPDATE Students
SET Mark = 65
WHERE StudentID = 104;
```

Important Use a primary key to identify the exact record to update or delete. A missing WHERE condition may change every record in the table.

9.5 Validation and database security

- Validation examples: data type check, range check, presence check, length check, format check, list/lookup check and uniqueness check.
- Use passwords, roles/permissions, encryption, audit trails, regular backups and controlled import/export to protect data.
- External data sources can be imported or linked. Check field types, formats and duplicate records after importing. A database connection lets another program or website exchange data with the DBMS securely.

10. WEB DESIGN

A website combines structured content, design, navigation and security.

10.1 Key terms

Term	Meaning
Web page	A single document viewed in a web browser.
Website	A collection of related web pages under one domain.
Browser	Software used to request and display web pages.
Web server	Computer/software that stores and serves web content.
HTML	HyperText Markup Language; structures page content.
CSS	Cascading Style Sheets; controls presentation and layout.
CMS	Content Management System; helps users create/manage website content, often using templates.
URL	Address used to locate a resource on the web.

10.2 Basic HTML page structure

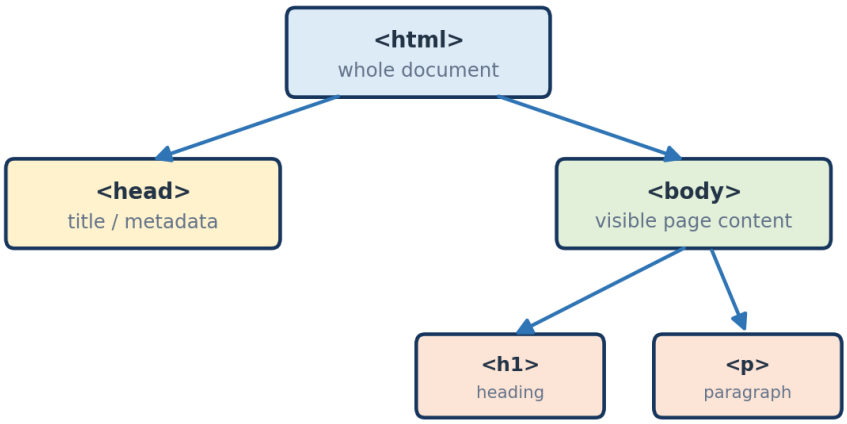


Figure 9: A simplified HTML document structure.

```
<!DOCTYPE html>
<html>
<head>
  <title>My School</title>
</head>
<body>
  <h1>Welcome</h1>
  <p>Welcome to our website.</p>
  <a href="about.html">About us</a>
</body>
</html>
```

Tag	Purpose
<html>	Root element of the page.
<head>	Contains metadata and page information.
<title>	Sets the browser tab title.
<body>	Contains visible page content.
<h1> to <h6>	Headings from highest to lower levels.
<p>	Paragraph.
<a>	Hyperlink.
<img>	Displays an image; include alternative text.
, ,	Unordered list, ordered list and list item.

10.3 Good web design

- Use consistent navigation, readable fonts, suitable contrast, clear headings, meaningful images and concise content.
- Menus and submenus organise pages. Links should work and images should have relevant alternative text.
- CMS templates speed up development. Plugins/extensions add features, but should be checked for reliability, updates and security.
- Web content may include text, images, audio, video, animation, web feeds and advertisements.

10.4 Web security and testing

- HTTPS encrypts data between browser and website. Use strong passwords, secure forms, software updates and trusted plugins.
- Cookies are small pieces of data stored by the browser for functions such as sessions and preferences; they can also be used for tracking.
- Browser developer tools help inspect page structure, styles, scripts, network requests and errors.
- Test navigation, links, forms, images, layout on different screen sizes, accessibility and browser compatibility.

11. TECHNOPRENEURSHIP

Technopreneurship applies technology and innovation to identify opportunities and create sustainable value.

11.1 Key concepts

- Entrepreneur: a person who organises a business and accepts calculated risk. Technopreneur: an entrepreneur whose product, service or business model uses technology or innovation.
- Innovation: introducing a new or improved product, service, process or business method that creates value.
- Market research: gathering information about customers, competitors, demand, prices and market conditions before launching a product.
- A business plan explains the business idea, target market, product/service, operations, marketing, resources, costs, finance and expected income. E-commerce uses digital networks to buy and sell goods or services.

11.2 Business start-up and growth

Area	What to consider
Opportunity	What problem or customer need will the product solve?
Target market	Who is most likely to buy/use it? Where are they?
Resources	Equipment, software, skills, staff, premises, Internet and suppliers.
Finance	Personal savings, loans, grants, investors, partners or reinvested profit.
Marketing	Product, price, place and promotion; use channels that reach the target market.
Risks	Competition, cash flow, technology failure, changing demand, regulation and cyber threats.

11.3 Technopreneurship support structures

- Science parks: support technology-based research, collaboration and commercialisation.
- Incubation centres: provide early-stage businesses with guidance, workspace, training, networks and sometimes access to funding.
- Academic institutions and research-and-development centres can provide expertise, research facilities and opportunities to develop innovations.

11.4 Intellectual property (IP)

Protection	What it protects (general meaning)
Copyright	Original creative expression such as software code, text, music, images and videos.
Patent	A qualifying new invention or technical solution, subject to legal requirements and registration.
Trademark	A brand identifier such as a name, logo or sign that distinguishes goods/services.
Trade secret	Confidential business information that provides a competitive advantage and is kept secret.

Ethical business practice Respect licences and creators' rights, protect customer data, advertise honestly, consider environmental impact and treat workers and customers fairly.

11.5 Intellectual capital, innovation and compliance

- Human capital: people's knowledge, skills and experience. Organisational capital: systems, processes, databases and knowledge held by the organisation. Social capital: useful relationships, networks and trust.
- Design thinking is a user-centred innovation approach: empathise with users, define the problem, generate ideas, build a prototype and test/improve it.
- Environmental technopreneurship develops solutions that reduce waste, pollution or resource use. Businesses must follow relevant laws, licences, policies and consumer/data-protection requirements.

12. QUICK-REVISION CHECKLIST

Use this page for the final day before the examination.

Can you do this without looking at your notes?	Tick
Name and explain the main input, output, processing and storage devices.	☐
Compare RAM, ROM, cache, HDD and SSD.	☐
Explain system, application and utility software.	☐
Give examples of computing in agriculture, health, banking, mining and transport.	☐
Convert between binary, denary, octal and hexadecimal.	☐
Add/subtract binary numbers and interpret ASCII.	☐
Complete truth tables for AND, OR, NOT, NAND, NOR and XOR.	☐
Draw and compare PAN, LAN, MAN, WAN and star/bus/ring/mesh topologies.	☐
Explain simplex, half-duplex and full-duplex.	☐
Identify network devices and common protocols.	☐
Explain confidentiality, integrity, availability, validation and verification.	☐
Describe SDLC stages and feasibility factors.	☐
Write/trace sequence, selection and loop algorithms.	☐
Identify syntax, runtime and logic errors; plan tests.	☐
Draw an ERD and write a simple SQL SELECT query.	☐
Create a basic HTML page and test its links.	☐
Explain business plans, funding and intellectual property.	☐

13. PRACTICE QUESTIONS

Attempt these questions before checking the answer guide. The questions are original revision practice, not copied examination questions.

Section A - Multiple choice

1. Which component performs arithmetic calculations and logical comparisons?
A. Control Unit B. ALU C. RAM D. Hard disk

2. Which memory loses its contents when the power is switched off?
A. ROM B. SSD C. RAM D. Optical disc

3. Which device is most suitable for producing a large engineering drawing?
A. Plotter B. Microphone C. Barcode reader D. Speaker

4. Which is an example of utility software?
A. Spreadsheet B. Antivirus C. Database table D. Presentation

5. What is the denary value of 101101_2?
A. 41 B. 43 C. 45 D. 49

6. What is AD_{16} in binary?
A. 10101101 B. 11011010 C. 10110101 D. 11101101

7. Which gate outputs 1 only when both inputs are 1?
A. OR B. XOR C. NOT D. AND

8. A network connecting a phone to wireless earbuds is usually a:
A. WAN B. PAN C. MAN D. LAN

9. Which device connects different networks and forwards packets?
A. Hub B. Bridge C. Router D. Repeater

10. Which transmission mode allows simultaneous two-way communication?
A. Simplex B. Half-duplex C. Full-duplex D. Broadcast

11. Which protocol translates domain names into IP addresses?
A. FTP B. DNS C. SMTP D. HTTP

12. Which measure scrambles data so unauthorised users cannot easily read it?
A. Compression B. Encryption C. Validation D. Formatting

13. Checking whether a mark is between 0 and 100 is a:
A. Verification check B. Range validation C. Authentication check D. Backup

14. Which feasibility type asks whether the organisation can afford the system?
A. Legal B. Operational C. Economic D. Technical

15. Which flowchart symbol represents a decision?
A. Rectangle B. Diamond C. Oval D. Parallelogram

16. Which programming construct repeats instructions?
A. Sequence B. Selection C. Iteration D. Assignment

17. Which error allows a program to run but gives the wrong answer?
A. Syntax B. Runtime C. Logic D. Translation

18. Which database field uniquely identifies each record?
A. Foreign key B. Primary key C. Query field D. Validation rule

19. Which database object is used to produce a formatted printable output?
A. Form B. Query C. Report D. Table

20. Which HTML tag creates a hyperlink?
A. <p> B. <img> C. <a> D. <h1>

21. Which organisation commonly supports early-stage businesses with advice and workspace?
A. Incubation centre B. ISP C. Router D. Database

22. Which intellectual property protection generally relates to an invention?
A. Copyright B. Patent C. Cookie D. Password

Section B - Structured questions

1. Hardware and software [6]

(a) State two differences between RAM and ROM. [2]
(b) Give two functions of an operating system. [2]
(c) Name one utility program and state its purpose. [2]

2. Applications and control systems [6]

A farmer installs an automated greenhouse system. (a) Name a suitable sensor for monitoring the greenhouse temperature. [1] (b) Explain the role of the controller. [2] (c) State two benefits of using an automated system. [2] (d) What is feedback? [1]

3. Data representation [8]

(a) Convert 110101_2 to denary. Show working. [2] (b) Convert 58_{10} to binary. [2] (c) Convert 111010_2 to octal. [2] (d) Calculate $1011_2 + 0110_2$. [2]

4. Networks [8]

(a) Distinguish between a LAN and a WAN. [2] (b) Draw a labelled star topology with four computers and a switch. [3] (c) State one advantage and one disadvantage of star topology. [2] (d) Name the device that connects different networks. [1]

5. Security and ethics [6]

(a) Explain phishing. [2] (b) State two ways of protecting a school database. [2] (c) Distinguish between validation and verification. [2]

6. Systems analysis and design [6]

(a) State three methods of collecting information about an existing system. [3] (b) What is a feasibility study? [1] (c) Name two areas of feasibility. [2]

7. Algorithms and programming [8]

Write pseudocode that inputs three test marks, calculates their average and displays “Pass” if the average is at least 50; otherwise display “Fail”. [8]

8. Databases [8]

A school database contains a STUDENTS table with StudentID, StudentName, ClassName and Mark. (a) Identify a suitable primary key. [1] (b) Define a record. [1] (c) Write an SQL query to display StudentName and Mark for students with a mark of 60 or more, sorted by mark from highest to lowest. [4] (d) State one purpose of a database form and one purpose of a report. [2]

9. Web design [5]

(a) State the purpose of HTML. [1] (b) Write the HTML tag used for a paragraph. [1] (c) State two features of a well-designed website. [2] (d) What is a CMS? [1]

10. Technopreneurship [5]

(a) Define technopreneurship. [2] (b) Name two possible sources of finance for a new technology business. [2] (c) State one purpose of copyright. [1]

14. ANSWER GUIDE

Check your answers only after attempting the practice questions.

Section A - Multiple choice

Q	Answer	Reason / reminder
1	B	ALU
2	C	RAM
3	A	Plotter
4	B	Antivirus
5	C	45
6	A	10101101
7	D	AND
8	B	PAN
9	C	Router
10	C	Full-duplex
11	B	DNS
12	B	Encryption
13	B	Range validation
14	C	Economic
15	B	Diamond
16	C	Iteration
17	C	Logic
18	B	Primary key
19	C	Report
20	C	<a>
21	A	Incubation centre
22	B	Patent

Section B - Suggested points

1. Hardware and software

(a) RAM is volatile; ROM is non-volatile. RAM is read/write working memory; ROM stores firmware/start-up instructions. (b) Any two: file, memory, device, processor, security or user-interface management. (c) Antivirus detects/removes malware; backup utility copies data; compression utility reduces file size; disk clean-up removes unnecessary files.

2. Applications and control systems

(a) Temperature sensor. (b) The controller receives the sensor reading, compares it with a set value and decides whether to activate an output such as a fan/heater. (c) Any two: maintains suitable conditions, reduces manual work, improves consistency, can operate continuously, records data. (d) Feedback is information about the actual output returned to the controller for comparison.

3. Data representation

(a) $110101_2 = 32 + 16 + 0 + 4 + 0 + 1 = 53_{10}$. (b) $58_{10} = 111010_2$. (c) $111010_2 = 111\ 010_2 = 72_8$. (d) $1011_2 + 0110_2 = 10001_2$.

4. Networks

(a) LAN covers a small area such as a school; WAN covers a large geographical area and may connect multiple LANs. (b) Four computers each connect by a separate cable to a central switch; label computers and switch. (c) Advantage: easy to add devices / one cable failure affects one device. Disadvantage: switch failure stops the network / needs more cable. (d) Router.

5. Security and ethics

(a) Phishing is a deceptive email/message/site designed to trick a person into disclosing passwords, bank details or other sensitive data. (b) Any two: strong passwords, access rights, encryption, firewall, antivirus, backups,

physical security, audit logs. (c) Validation checks data is sensible/within set rules; verification checks data has been entered/copied accurately.

6. Systems analysis and design

(a) Questionnaire, interview, observation, document/record inspection. (b) A feasibility study evaluates whether a proposed system is practical and worth implementing. (c) Any two: technical, economic, legal, operational, social.

7. Algorithms and programming

One valid answer: START; total = 0; FOR counter = 1 TO 3; READ mark; total = total + mark; NEXT counter; average = total / 3; DISPLAY average; IF average >= 50 THEN DISPLAY "Pass" ELSE DISPLAY "Fail" ENDIF; STOP. Award credit for correct input, initialisation, loop, accumulation, average, decision and output.

8. Databases

(a) StudentID. (b) A record is a complete set of fields about one entity, usually represented by a row in a table. (c) SELECT StudentName, Mark FROM Students WHERE Mark >= 60 ORDER BY Mark DESC; (d) A form makes data entry/viewing easier; a report formats data for presentation or printing.

9. Web design

(a) HTML structures the content of a web page. (b) <p>. (c) Any two: clear navigation, readable text, consistent layout, suitable contrast, working links, responsive design, accessible images/content. (d) A Content Management System is software used to create, edit and manage website content, often using templates.

10. Technopreneurship

(a) Applying technology and innovation to create or grow a business that meets a need or solves a problem. (b) Any two: savings, bank loan, grant, investor, partner, family contribution or reinvested profits. (c) Copyright gives creators legal protection over original creative work and controls specified forms of copying/use.

15. LAST-MINUTE EXAM TECHNIQUE

Small habits that help candidates earn marks.

- Read the command word: state/name = short answer; describe = give characteristics; explain = give a point and why/how; compare = show similarities/differences; draw = use correct symbols and labels.
- For calculations, show the method and the final answer with the correct base/unit.
- For diagrams, use a ruler where appropriate, label devices/links/symbols and show the direction of data/control flow.
- For pseudocode, initialise variables, use meaningful names, indent blocks and close IF/loop structures correctly.
- For SQL, check field and table names, quotation marks around text, WHERE conditions and ORDER BY direction.
- For practical work, save frequently, use meaningful filenames, test each feature and follow the submission instructions given by the centre.

Candidate reminder Practise with a computer as well as reading notes. The practical paper tests what you can create, run, test and present - not only what you can define.

SYLLABUS AND EXAMINATION REFERENCES

- Ministry of Primary and Secondary Education, Zimbabwe. Computer Science Syllabus, Forms 1-4, 2024-2030 (Heritage-Based Curriculum). https://studylib.net/doc/28527371/computer-science-o-level-syllabus
- ZIMSEC. Computer Science 4021/02 - 2026 specimen paper (official ZIMSEC download page). https://www5.zimsec.co.zw/download/computer-science-4021-02/
- ZIMSEC Computer Science 4021/1 - 2026 specimen paper (multiple choice format and syllabus coverage). https://zimsec.invalcre.co.zw/past-papers/o-level/o-level-computer-science/paper-1/specimen-2026/
- ZIMSEC Computer Science 4021/3 - 2026 specimen paper (practical format: programming, databases and web design). https://sytbay.co.zw/downloads/zimsec-o-level-computer-science-specimen-papers-4021-1-4021-2-4021-3-2026-hbc-pdf/

Note: The practice questions in this booklet are original revision exercises written to cover the syllabus areas. Candidates should follow the exact instructions and paper arrangements issued by ZIMSEC and their examination centre.